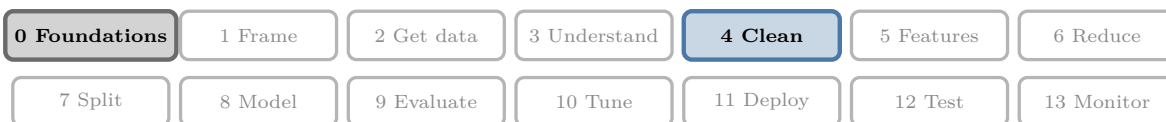


The Standard Normal Distribution and the Z-table

Contents

1. Overview	2
2. The standard normal distribution	3
3. Standardizing a normal variable	3
3.1 Why standardize?	4
4. The z-table	4
5. Two problems solved with the z-table	5
5.1 Taller than 72 inches	5
5.2 Between the mean and one standard deviation above it	6
6. Deriving the 68-95-99.7 rule	7
7. Where the normal distribution is used in data science	7
8. Summary	9
Sources	9
9. Key terms	9

Where this fits: Pipeline map steps 0 (Foundations), 4 (Clean). See the [Course map](#).



- **Builds on:** Skewness ([Note 20](#)); Standardization ([Note 24](#)); Descriptive statistics ([Note 230](#)); Cumulative distribution function (CDF) ([Note 242](#)); Normal distribution ([Note 250](#)).
- **Leads to:** Confidence intervals ([Note 280](#)); Z-test and rejection regions ([Note 291](#)).
- **Compare with:** IQR outlier method ([Note 43](#)); Percentile outlier method ([Note 44](#)).

1. Overview

Key point: Standardizing turns any normal distribution into the standard normal distribution $N(0, 1)$; one table of its areas, the z-table, then answers probability questions about every normal variable.

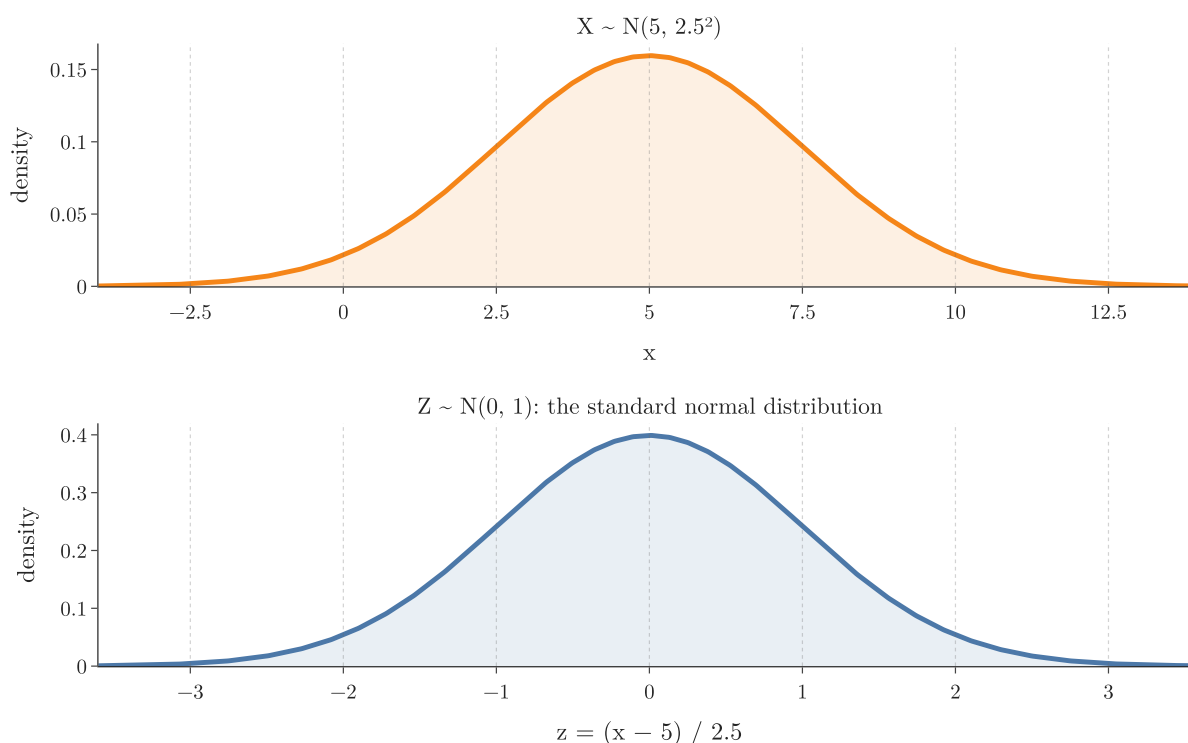


Figure 1: Standardizing: $X \sim N(5, 2.5^2)$ becomes $Z \sim N(0, 1)$; every value moves to its z-score

Figure 1 shows the idea. The orange curve is a normal distribution with mean 5 and standard deviation 2.5. Subtracting 5 and dividing by 2.5 turns it into the blue curve, which has mean

0 and standard deviation 1. The shape stays exactly the same; only the numbers on the axis change.

This Note builds on the [normal distribution Note](#). It introduces the standard normal distribution, shows how to read a z-table, uses it on two problems, derives the 68-95-99.7 rule, and ends with where the normal distribution is used in data science.

2. The standard normal distribution

Key point: The standard normal distribution is the normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0, 1)$.

Among all normal distributions, one is special: the one with $\mu = 0$ and $\sigma = 1$. It is called the **standard normal distribution** (or **standard normal variate**) and is written with the letter Z :

$$Z \sim N(0, 1)$$

Its curve (Figure 1, bottom) is centred at 0, and its x axis counts standard deviations directly: 1 means one standard deviation above the mean, -2 two below.

1. **In words:** put $\mu = 0$ and $\sigma = 1$ into the normal PDF.
2. **Formula:**

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

The standard normal PDF has its own symbol, ϕ (phi), and its CDF is written Φ (capital phi).

3. **Example:** at $z = 0$, $\phi(0) = 1/\sqrt{2\pi} = 1/2.5066 = 0.3989$, the height of the peak in Figure 1. At $z = 1$, $\phi(1) = 0.3989 \times e^{-0.5} = 0.3989 \times 0.6065 = 0.2420$.

3. Standardizing a normal variable

Key point: Subtracting the mean and dividing by the standard deviation turns $X \sim N(\mu, \sigma^2)$ into $Z \sim N(0, 1)$.

To turn any normal variable into a standard normal one, we standardize every value: subtract the mean, divide by the standard deviation. The result is its z-score. This is the standardization of the [standardization Note](#):

$$z = \frac{x - \mu}{\sigma}$$

For Figure 1, $x = 10$ becomes $z = (10 - 5)/2.5 = 2$: the value 10 lies two standard deviations above the mean. Every tick on the top axis lines up with its z-score below.

Real data works the same way. The **Age** column of the Titanic data (714 known ages) is roughly bell-shaped, with mean 29.70 years and standard deviation 14.53 years. Standardizing it gives a column with mean 2×10^{-16} (0 up to rounding) and standard deviation 1.

Python: Standardizing a column by hand.

```
import pandas as pd
```

```
df = pd.read_csv("data/titanic_train.csv")
```

```
age = df["Age"].dropna()
```

```
z = (age - age.mean()) / age.std()
```

```
z.mean(), z.std()          # about 0, and 1
```

Standardizing changes the scale, not the shape: a skewed column stays skewed (see the [standardization Note](#), section 7.3). Only a normal column becomes standard normal.

3.1 Why standardize?

Key point: Standardizing lets us compare different normal distributions on one scale, and compute every probability from one table.

Two benefits make the standard normal distribution so useful:

1. **Comparison.** Two normal distributions with different means and spreads (heights in centimetres and in inches, marks in two different exams) can be compared side by side once both are standardized.
2. **One table for all probabilities.** The areas under the standard normal curve have been computed once and printed in a table. Any probability about any normal variable becomes a lookup in that table after standardizing.

4. The z-table

Key point: A z-table lists $\Phi(z)$, the area under the standard normal curve to the left of z : the probability that Z is at most z .

A **z-table** (standard normal table) lists the CDF of the standard normal distribution for many values of z :

$$\Phi(z) = P(Z \leq z) = \text{area under the curve from } -\infty \text{ to } z$$

The table gives an **area**, so a probability, not the height of the curve at z (Figure 2, left). There are two tables: one for negative z and one for positive z .

To look up a z-score with two decimals, such as 1.33:

1. Choose the positive table, because 1.33 is positive.
2. Find the row for the first two digits, **1.3**.
3. Find the column for the second decimal, **0.03**.
4. Read the cell where they cross: **0.90824** (Figure 2, right).

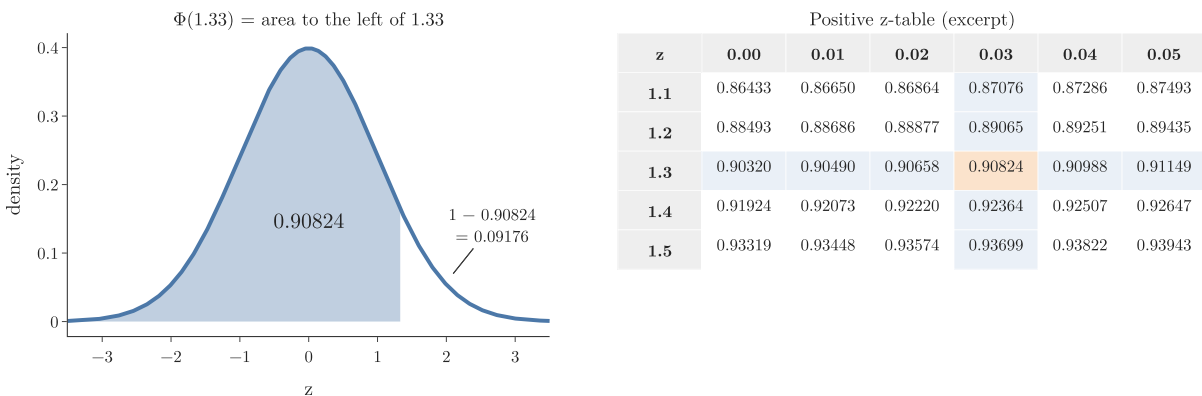


Figure 2: The area to the left of $z = 1.33$ (left) and where it sits in the z-table (right)

So $P(Z \leq 1.33) = 0.90824$: 90.8% of the area lies to the left of 1.33. The area to the right is the rest, $1 - 0.90824 = 0.09176$.

Python: The z-table in one line.

```
from scipy import stats
```

```
stats.norm.cdf(1.33)    # 0.90824, Phi(1.33)
stats.norm.sf(1.33)     # 0.09176, 1 - Phi(1.33)
stats.norm.ppf(0.975)   # 1.96, the z with 97.5% to its left
```

5. Two problems solved with the z-table

Key point: Standardize the value, look up its area, and subtract when the question asks for “more than” or “between”.

5.1 Taller than 72 inches

Key point: About 9.2% of men are taller than 72 inches when heights are $N(68, 3^2)$.

The heights of adult men in a population are normal with mean 68 inches and standard deviation 3 inches. What is the probability that a randomly chosen man is taller than 72 inches?

1. **In words:** turn 72 into a z-score, read the area to its left, and subtract it from 1 to get the area to its right.
2. **Formula:**

$$P(X > 72) = 1 - \Phi\left(\frac{72 - 68}{3}\right)$$

3. **Example:** the z-score is

$$z = \frac{72 - 68}{3} = \frac{4}{3} = 1.33$$

The z-table gives $\Phi(1.33) = 0.90824$, so

$$P(X > 72) = 1 - 0.90824 = 0.09176$$

About 9.2% of men are taller than 72 inches. (With the unrounded $z = 1.3333$, software gives 0.0912.)

The steps are always the same: draw the curve, standardize the value, look up the area, then subtract if needed.

5.2 Between the mean and one standard deviation above it

Key point: For every normal distribution, 34.13% of the values lie between μ and $\mu + \sigma$.

Now no numbers are given: $X \sim N(\mu, \sigma^2)$. What share of the values lies between the mean μ and $\mu + \sigma$?

1. **In words:** standardize both ends, then subtract the area left of the lower end from the area left of the upper end.
2. **Formula:**

$$z_{\text{lower}} = \frac{\mu - \mu}{\sigma} = 0, \quad z_{\text{upper}} = \frac{(\mu + \sigma) - \mu}{\sigma} = 1$$

$$P(\mu \leq X \leq \mu + \sigma) = \Phi(1) - \Phi(0)$$

3. **Example:** the z-table gives $\Phi(1) = 0.8413$, and $\Phi(0) = 0.5$ because the curve is symmetric. So

$$P(\mu \leq X \leq \mu + \sigma) = 0.8413 - 0.5 = 0.3413$$

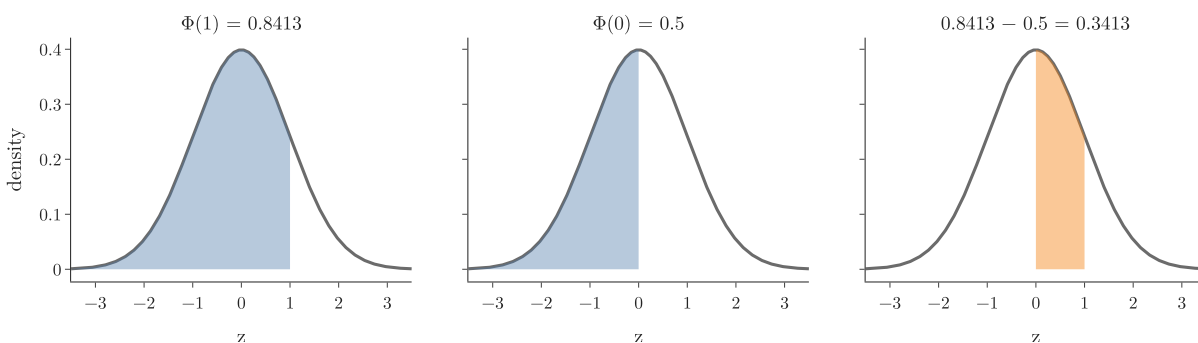


Figure 3: The area between 0 and 1 as the difference of two z-table areas

The answer, 34.13%, does not depend on μ or σ : it holds for every normal distribution.

6. Deriving the 68-95-99.7 rule

Key point: Repeating the calculation for 1, 2 and 3 standard deviations, and doubling by symmetry, gives 68.27%, 95.45% and 99.73%.

By symmetry, the same 34.13% lies between $\mu - \sigma$ and μ . So the share within one standard deviation on either side is

$$P(\mu - \sigma \leq X \leq \mu + \sigma) = 2 \times 0.3413 = 0.6827$$

The same steps for 2 and 3 standard deviations:

Range	Area right of the mean	Both sides
$\mu \pm 1\sigma$	$\Phi(1) - 0.5 = 0.8413 - 0.5 = 0.3413$	68.27%
$\mu \pm 2\sigma$	$\Phi(2) - 0.5 = 0.9772 - 0.5 = 0.4772$	95.45%
$\mu \pm 3\sigma$	$\Phi(3) - 0.5 = 0.99865 - 0.5 = 0.49865$	99.73%

This is the **empirical rule** of the [z-score outliers Note](#), now derived from the z-table instead of taken on trust. It is powerful: knowing only that a variable is normal, without seeing any data, we can say that 99.73% of its values lie within 3 standard deviations of the mean.

A value far outside that range is extraordinary. Don Bradman's Test batting average of 99.94 is a famous example: it lies about 4.4 standard deviations above the mean of Test cricketers (Davis 2000). If the averages were normal, fewer than 1 value in 100,000 would lie that far up: $1 - \Phi(4.4) \approx 0.000005$. Roughly speaking, most good batsmen sit within one or two standard deviations of the mean, and only the very greatest approach three.

7. Where the normal distribution is used in data science

Key point: Outlier detection, the assumptions of some ML models, hypothesis testing, and the central limit theorem.

1. **Outlier detection.** For a column that is roughly normal, values beyond $\mu \pm 3\sigma$ are treated as outliers (the z-score method, see the [z-score outliers Note](#)). For the Titanic ages the limits are $29.70 \pm 3 \times 14.53$, from -13.88 to 73.28 years. No age is negative, so only the upper limit matters: two passengers, aged 74 and 80, are flagged.
2. **Assumptions of ML models.** Some models perform better, or rely on the assumption, that something is normally distributed. Linear regression assumes that the **residuals** (the errors) are normal, not the inputs (see the [linear regression assumptions Note](#)). Linear and logistic regression also tend to work better on normal-looking inputs (see the [function transformer Note](#)), and a Gaussian mixture model is built from normal curves (see the [density estimation Note](#)).

3. **Hypothesis testing.** Many statistical tests assume that the data, or a statistic computed from it, is normally distributed.
4. **The central limit theorem.** Averages of samples from almost any distribution, normal or not, follow approximately a normal distribution, more closely as the samples grow (Pishro-Nik §7.1.2). This result, the topic of a later Note, is what makes the normal distribution central to inferential statistics.

8. Summary

Idea	Formula	Example
Standard normal	$Z \sim N(0, 1)$	peak $\phi(0) = 0.3989$
Standardize	$z = (x - \mu)/\sigma$	$(72 - 68)/3 = 1.33$
Z-table	$\Phi(z) = P(Z \leq z)$	$\Phi(1.33) = 0.90824$
More than x	$1 - \Phi(z)$	$P(X > 72) = 0.092$
Between two values	$\Phi(z_2) - \Phi(z_1)$	$\Phi(1) - \Phi(0) = 0.3413$
Empirical rule	$2(\Phi(k) - 0.5)$	68.27%, 95.45%, 99.73%

- A z-table gives areas (probabilities) to the left of z , not heights of the curve.
- Standardizing keeps the shape; only normal data becomes standard normal.
- Values beyond 3 standard deviations are rare: 0.27% in all.

Sources

- Davis, C. (2000). *The Best of the Best*. Bradman's 4.4 standard deviations as reported in *Discover* magazine, "Who's the Greatest Sports person?"
- Pishro-Nik, H. (2014). *Introduction to Probability, Statistics, and Random Processes*. Kappa Research. Section 7.1.2 (central limit theorem; needs a finite variance).

9. Key terms

Term	Meaning
Standard normal distribution	The normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0, 1)$
$\phi(z)$	The PDF of the standard normal distribution
$\Phi(z)$	The CDF of the standard normal distribution: the area to the left of z
Z-table	A table of $\Phi(z)$ for many values of z , used to find normal probabilities